

Yelp Sentiment Analysis Case Study Rubric

Submission format: Upload a link of your Github page to canvas.

General Description: Complete the Yelp Sentiment Analysis case study by engaging with the provided materials in the GitHub repository and producing a well-documented notebook. Your work should demonstrate a clear understanding of how text data can be analyzed to understand customer sentiment.

Why am I doing this? This case study is designed to help you apply the core concepts of data science to a real-world problem: understanding customer sentiment from Yelp restaurant reviews. You will practice working with unstructured text data and learn how to extract meaningful insights from it.

By completing this case study, you will:

- Gain experience analyzing text data
- Understand how machine learning models can be applied to real-world datasets
- Practice communicating your findings clearly

This assignment is also an opportunity to develop your ability to explain technical work to a broader audience.

What am I going to do? In this case study, you will analyze Yelp restaurant review data to explore how written reviews relate to customer ratings.

You will:

- Explore the dataset and identify key variables
- Perform basic data cleaning and preparation
- Conduct exploratory data analysis (EDA) to identify patterns in the data
- Convert text data into numerical features (like TF-IDF)
- Build a classification model to predict sentiment (positive vs. negative)
- Evaluate the model and interpret the results

Your goal is not to build the most complex model, but to clearly demonstrate your understanding of the data science process and explain your findings.

Deliverables include:

- A completed notebook that walks through your analysis
- Clear visualizations and explanations of your findings
- A working model that predicts sentiment from text
- Interpretation of model results in plain language

Tips for success:

- Be clear and intentional. Focus on explaining your reasoning at each step.
- Don't overthink it. This is about understanding fundamentals, not advanced techniques.
- Keep your notebook organized and easy to follow.
- Focus on interpretation. What do your results actually mean?
- Think about your audience. Your explanations should be understandable to another student.

How will I know I have Succeeded? You will meet expectations on the case study when you follow the criteria in the rubric below.

Formatting	• One GitHub repository (submitted via link on Canvas) • Create a new GitHub repository for this assignment that contains: ◦ README.md ◦ Notebook file(s) (EDA and model) ◦ Data file(s) used for analysis ◦ Output files (if applicable, such as selected EDA or model visuals)
README.md	• Brief summary of what you have produced for the case study • Should explain: ◦ The purpose of the analysis ◦ What the notebook contains ◦ How to run or navigate the project • This does not need to be overly detailed, but should provide enough information to orient the reader
Notebook File	Well-documented Jupyter Notebook or Google Colab notebook that contains the code used to perform your analysis. The notebook should include: • Data loading and inspection • Creation of a sentiment label (positive vs. negative) • Basic text cleaning and preprocessing • Exploratory data analysis (visualizations and interpretation) • Feature engineering (TF-IDF or similar method) • A classification model • Model evaluation and interpretation of results • Comments and markdown explanations

	throughout ○ Explanations should describe what is being done and why ○ Interpret results in plain language
Data	● Include the dataset used for analysis ● Ensure the data is accessible and usable by someone viewing the repository